Beyond the Rate: Information Content of FOMC Forward Guidance Language

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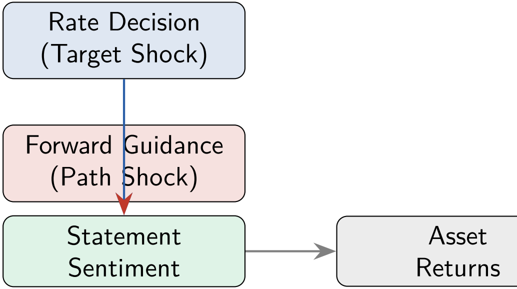
Outline

The Puzzle

Central question:
Does FOMC statement language convey information beyond the rate decision?

Why it matters:

- Central banks increasingly rely on **forward guidance** as a policy tool
- Markets respond to *what the Fed says*, not just *what it does*
- The "information channel" (Romer & Romer, 2000) suggests FOMC statements reveal the Fed's superior economic assessment



Literature

- **Kuttner (2001)**: 30-min FF futures window $\rightarrow$ isolate monetary policy surprises
- **GSS (2005a)**: Decompose surprises into target + path factors
- **Nakamura & Steinsson (2018)**: High-frequency identification of monetary non-neutrality
- **Campbell et al. (2012)**: Delphic vs. Odyssean forward guidance
- **Jarociński & Karadi (2020)**: Separate policy shock from information shock
- **Hansen et al. (2018)**: Computational linguistics for FOMC transparency

Our contribution:

1. First to directly link **GSS target/path shocks** to **FOMC statement sentiment**
2. Expanded CB dictionary (97 hawkish + 106 dovish terms, fully disjoint)
3. Formal test of the **information channel** via text analysis

Data

Monetary Policy Shocks

- Acosta (2022): 220 FOMC meetings, 1995–2022
- Target shock: surprise in current FF rate
- Path shock: surprise in future rate path
- Extension: USMPD (SF Fed) $\rightarrow$ 2026

Market Data

- CRSP VW/EW/S&P 500 via WRDS
- 117 meetings with complete overlap

FOMC Statement Sentiment

- 164 statements, 2006–2025
- Expanded CB dictionary:
  - 97 hawkish terms
  - 106 dovish terms
  - 50 bigram phrases
  - Fully disjoint
- Score $= \frac{H-D}{H+D+\epsilon}$
- Validated: $r = 0.29$ with rate change

Empirical Strategy

Four hypotheses:

Hypothesis	Prediction	Specification
H1	Shocks predict sentiment	$S_t = \alpha + \beta_1 \text{Target}_t + \beta_2 \text{Path}_t + \varepsilon_t$
H2	Shocks predict returns	$R_t = \alpha + \beta_1 \text{Target}_t + \beta_2 \text{Path}_t + \varepsilon_t$
H3	Path $>$ Target for sentiment	$\lvert\beta_2\rvert > \lvert\beta_1\rvert$ in H1
H4	Stronger during FG period	Interaction with FG dummy

Estimation: OLS with Newey-West (lag=4, data-driven) standard errors

Sample: 117 FOMC meetings, January 2006 – July 2022

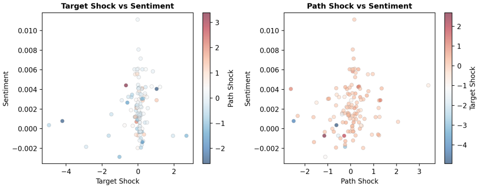
H1: Sentiment and Monetary Policy Shocks

	Coefficient	p-value
Target shock	0.000290	0.062*
Path shock	0.000469	0.047**
R^2	4.06%	
N	117	

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$

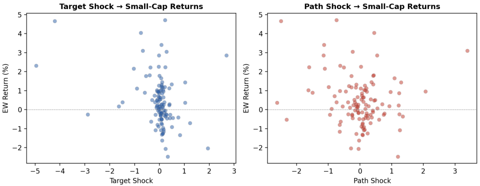
Key finding: Path shock is significant at 5%, target at 10%.

But: Wald test $\chi^2 = 0.19$, $p = 0.66$ — cannot reject $\beta_1 = \beta_2$.



H2: Asset Returns and Shocks

Asset	β_T	p_T	β_P	p_P
CRSP VW	-0.435	0.111	-0.186	0.398
CRSP EW	-0.449	0.044**	-0.174	0.421
S&P 500	-0.391	0.158	-0.179	0.395
Returns in percentage points; $N = 117$				



Findings:

- Target shock: significant for EW (small-cap), not VW
- Path shock: not significant for any asset
- Small-cap sensitivity consistent with literature

H3: The Information Channel

Prediction: Path shock has larger effect on sentiment than target shock.

- $|t_{\text{path}}| = 2.012 > |t_{\text{target}}| = 1.887$
- Path significant at 5%, target at 10%
- But: Wald test $H_0 : \beta_{\text{target}} = \beta_{\text{path}}$
    - $\chi^2 = 0.19$, $p = 0.66$
    - Cannot reject equality

Interpretation: *Suggestive* evidence consistent with the information channel, but **not definitive proof** that the path effect dominates.

Why the modest R^2 (4.06%):

- 96% of sentiment variation is unexplained by shocks
- Likely reflects: economic data releases, institutional inertia, other Fed communication channels

Robustness Checks

Sample	R^2	p(Target)	p(Path)	N	Period
Full (Acosta)	4.06%	0.062^{*}	0.047^{**}	117	2006–22
No COVID	4.10%	0.075^{*}	0.042^{**}	115	2006–22
Post-2010	2.02%	0.369	0.108	97	2010–22
Extended	1.65%	0.419	0.058^{*}	163	2006–26

Key takeaways:

- COVID exclusion: virtually unchanged $\rightarrow$ not driven by outliers
- Post-2010: neither shock significant $\rightarrow$ ZLB period weakens the signal
- Extended sample (USMPD): path still significant at 10% through 2026

Regime-Specific Results

Period	N	R^2	p(Target)	p(Path)
Pre-ZLB (2006–08)	8	20.5%	0.360	0.403
Crisis (2008–10)	12	9.2%	0.930	0.124
ZLB/FG (2010–16)	48	7.5%	0.249	0.107
Normalization (2016–20)	30	13.3%	0.009***	0.873
COVID+ (2020–22)	19	3.6%	0.355	0.953

Caution: small N within each regime. Results are suggestive.

Pattern: Target dominates during Normalization (rate hikes); Path is more relevant during ZLB/FG and Crisis periods.

Limitations & Future Work

Limitations:

1. **Low R^2**: Shocks explain only 4% of sentiment variation
2. **No control variables**: No lagged sentiment, macro controls, or meeting type dummies
3. **Dictionary-based sentiment**: Cannot capture context or pragmatics
4. **Sample size**: $N = 117$ limits power, especially for sub-sample analysis
5. **US only**: External validity to other central banks unknown
6. **Multiple testing**: 4 hypotheses without significance level adjustment

Future directions:

- FinBERT / LLM-based sentiment for better measurement
- Control variables: lagged sentiment, macro indicators
- Cross-country: ECB, BoJ, BoE
- Other channels: minutes, press conferences, speeches

Conclusion

1. The **path shock** (forward guidance) is a statistically significant predictor of FOMC statement sentiment ($p = 0.047$)
2. The **target shock** is also significant at 10% ($p = 0.062$)
3. However, we **cannot formally establish** that the path effect dominates (Wald test $p = 0.66$)
4. The evidence is **suggestive** of an information channel, not definitive
5. Asset returns respond primarily to the **target shock** (small-cap sensitivity), not the path shock
6. Results are robust to COVID exclusion and sample extension (USMPD through 2026)

Takeaway: FOMC language is forward-looking, but monetary policy shocks explain only a small fraction of its variation. The information channel exists but is one of many forces shaping central bank communication.

Thank You

Questions & Discussion

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Appendix: Event Study Methodology

Market Model (Brown & Warner, 1985):

1. Estimation: $R_{it} = \alpha_i + \beta_i R_{mt} + \varepsilon_{it}$
2. Abnormal Return: $AR_{it} = R_{it} - (\hat{\alpha}_i + \hat{\beta}_i R_{mt})$
3. CAR: $CAR_i = \sum_{t=-T_1}^{T_2} AR_{it}$
4. **t-stat**: $t = \dfrac{CAR_i}{\sigma_{AR} \times \sqrt{N}}$

Important: The denominator is $\sigma_{AR} \times \sqrt{N}$, **not** $\sigma_{AR}/\sqrt{N}$.

- σ_{AR}: std. dev. of abnormal returns from estimation window
- N: number of days in event window
- Common error: using $\sigma_{AR}/\sqrt{N}$ inflates t by factor of N

For cross-sectional tests (average CAR across K assets), use $\sigma_{CAAR}/\sqrt{K}$.

Appendix: USMPD Factor Replication

GSS (2005a) two-step orthogonalized PCA:

1. Target = 1st PC of [MP1, FF1–FF4]
2. Path = 1st PC of [FF4–FF6, ED2–ED4], orthogonalized against target
3. Normalized by regressing on daily 1Y GSW yield change

Factor	Instruments	PC1 Variance	Corr w/ Acosta
Target	MP1, FF1–FF4	87.1%	0.958
Path (orth)	FF4–FF6, ED2–ED4	77.0%	0.970
STMT (single)	MP1, MP2, ED2–ED4	78.5%	0.989 (target+path)

Single-factor STMT replicates perfectly ($r = 1.000$).